

Securing Autonomous Driving Against Compound Sensory Attacks and Adversarial Dynamics: A Variance-Adaptive and Diversity-Regularized Defense Framework

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ABSTRACT

Deep reinforcement learning for autonomous driving is highly vulnerable to sensory perturbations and adversarial multi-agent dynamics. These vulnerabilities are particularly critical in intelligent transportation systems, where malicious sensor spoofing and coordinated adversarial agent attacks can directly compromise vehicle safety. Existing DRL controllers frequently suffer from severe value estimation degradation when subjected to such heterogeneous observation corruption and non-stationary adversarial interactions. To neutralize these compounded security threats, we propose a robust defense framework designed to stabilize policy optimization against joint observation-dynamics attacks. Specifically, we introduce a Variance-Adaptive Generalized Advantage Estimation mechanism that dynamically calibrates the bias-variance trade-off based on critic value variance, thereby insulating the value network from maliciously corrupted sensory inputs. Furthermore, we develop a hierarchical diversity regularizer to enforce structured behavioral dispersion, preventing policy overfitting against specific adversarial attack trajectories. Comprehensive evaluations within the MetaDrive simulator, encompassing six rigorous perturbation and threat regimes, demonstrate the superior robustness of our defense. The proposed framework significantly outperforms state-of-the-art baselines by improving the average episodic return by 34.1\% and increasing the safety-critical success rate by 14.9 percentage points, validating its efficacy in securing autonomous navigation against severe sensing and dynamic attacks.